

Macro-Financial Determinants of the Philippine Stock Exchange Index (PSEi) Returns

A Multiple Linear Regression Analysis 2018-2025

Presented by :
MACAYAN, Adrian Denzel
MIRANDA, Lara Stephanie
OLACO, Tiarra Emilia
ROA, Marco Antonio

CASE STUDY PRESENTATION 1

Section: **C02**



WHY STUDY PSEi RETURNS?



KEY RESEARCH QUESTION

Which domestic and global macro-financial factors are significantly associated with Philippine stock market returns and the risk of sharp monthly declines?

1. Market Uncertainty

Philippine stock returns are influenced by domestic economic conditions and global financial developments.

2. Research Gap

The strength and direction of these relationships may change under different economic conditions.

3. Study Focus

Examine how nine macro-financial indicators are associated with monthly PSEi returns from 2018 to 2025.

4. Methodology

Apply multiple linear regression and logistic regression to evaluate market relationships and downside risk.

5. Practical Relevance:

Help investors identify important economic indicators for portfolio monitoring and risk management.



DATA & VARIABLES



Philippine Stock Exchange Index (PSEi)

Dependent Variable

GLOBAL FACTORS

SP500 Return • WTI Return • VIX Index

DOMESTIC MACRO

PH_CPI • PH_M3 • PH_Manufacturing_Production

MONETARY & FX

BSP_RRP_Rate • PH_10Y_GS_Yield • USD_PHP



DATA PREPROCESSING

Harmonization & Variable Transformations

96 *MONTHLY OBSERVATIONS*

10 *TOTAL VARIABLES*

01/2018 *Starting Month* - **12/2025** *Ending Month*

Variable	Role	Original Frequency	Applied Transformation	Unit of Measurement
PSEi	Dependent Variable	Daily Trading	Monthly Log Return	Decimal (0.01 = 1%)
SP500 / WTI / USD_PHP	Regressors	Daily Trading	Monthly Log Return / Log Change	Decimal (0.01 = 1%)
BSP_RRP_Rate / PH_10Y_GS_Yield	Regressors	Event-dated / Daily	Month-end First Difference	Percentage Points
PH_CPI / PH_M3 / PH_Mfg_Prod	Regressors	Monthly	YoY % Change / Growth (Unchanged)	Percent (%)
VIX	Regressor	Daily Trading	Monthly Average (Unchanged level)	Index Points

FREQUENCY ALIGNMENT

Align mixed daily, dated event, and monthly series to standard calendar months.

Daily Series: *Full date join (2,109 dates)*
LOCF (tidyr::fill): *Carry forward missing market dates*
Policy Rates: *Latest rate prior to month-end*

MONTHLY AGGREGATION

Collapse daily observations into single monthly entries using tailored logic.

Prices/Yields: *Last valid day of calendar month*
VIX Volatility: *Monthly average across active trading days*
Base Month: *Dec 2017 retained for Jan 2018 initial metrics*

STATIONARY TRANSFORMATIONS

Convert persistent levels to stationary returns or differences to avoid spurious regression.

Log Returns: *PSEi, SP500, USD/PHP, WTI*
1st Differences: *BSP RRP & 10Y GS Yields*
Levels Retained: *CPI, M3, Mfg Growth & VIX*



DATA PREPROCESSING

Dataset Diagnostics & Quality Assurance



STRUCTURAL INTEGRITY

- ***Strict Date Alignment:*** *Exactly 96 contiguous months from Jan 2018 to Dec 2025.*
- ***Sorting Verified:*** *Chronologically ordered with zero missing calendar months.*
- ***No Imputation:*** *Zero rows removed or synthetically filled in final analysis matrix.*

TYPE & DISPERSION CHECKS

- ***Numeric Consistency:*** *All 10 series accurately parsed as continuous numeric variables.*
- ***Discrete Step Dynamics:*** *BSP RRP Rate yields 6 distinct values (0.25bp shifts; zero in 65/96 months).*
- ***No Degenerate Columns:*** *No constant series detected across the evaluation window.*

MACROECONOMIC OUTLIERS

- ***Pandemic Volatility:*** *Mfg Production ranges -81% to +498% (2020–2021 shock/rebound).*
- ***Zero Outlier Trimming:*** *Retained to test model resilience under structural stress.*
- ***Downstream Audit:*** *Influential observations evaluated via diagnostics in Section 7.*



Key Data Concerns

14
Mfg Observations
Outside -50% to +100% range

3
Extreme Returns
|log return| > 0.25

4
Mfg Outliers (|z|>3)
Standard z-score threshold

0
VIX / CPI / M3 Flags
No plausibility check flags

Variable	Mean	Median	Std. Dev.	Skewness	Ex. Kurtosis
PSEI	—	—	—	-0.94	1.99
VIX	19.73	18.34	6.76	2.37	9.50
WTI	—	—	—	-1.11	13.86
BSP_RRP_Rate	—	0.00	—	—	—
PH_Mfg_Prod	19.74	0.47	97.69	3.71	13.99

EXPLORATORY DATA ANALYSIS

Descriptive Statistics and Plausibility Checks

Central Tendency

- *Most variables are centered near zero.*
- *PH_Manufacturing_Production has a substantial mean-median difference (19.74 vs. 0.47).*
- *VIX also exhibits right-skewness (mean: 19.73; median: 18.34).*

Distribution Shape

- *Most variables are relatively symmetric.*
- *WTI, VIX, and PH_Manufacturing_Production exhibit heavy tails.*
- *PH_Manufacturing_Production has the highest positive skewness (3.71).*

Variance / Dispersion

- *PH_Manufacturing_Production has the highest variability (SD = 97.69).*
- *VIX also exhibits considerable dispersion (SD = 6.76).*
- *USD_PHP and SP500 have relatively low variability (SD = 0.016 and 0.048).*



EXPLORATORY DATA ANALYSIS

Time-Series Analysis

1. Global Market Disruptions

COVID-19 triggered sharp declines in SP500 and WTI, while VIX peaked at approximately 58.

2. Manufacturing Volatility

PH_Manufacturing_Production fell to -80.99% in 2020 before recording unusually high recovery growth in 2021–202

3. Delayed Inflationary Pressures

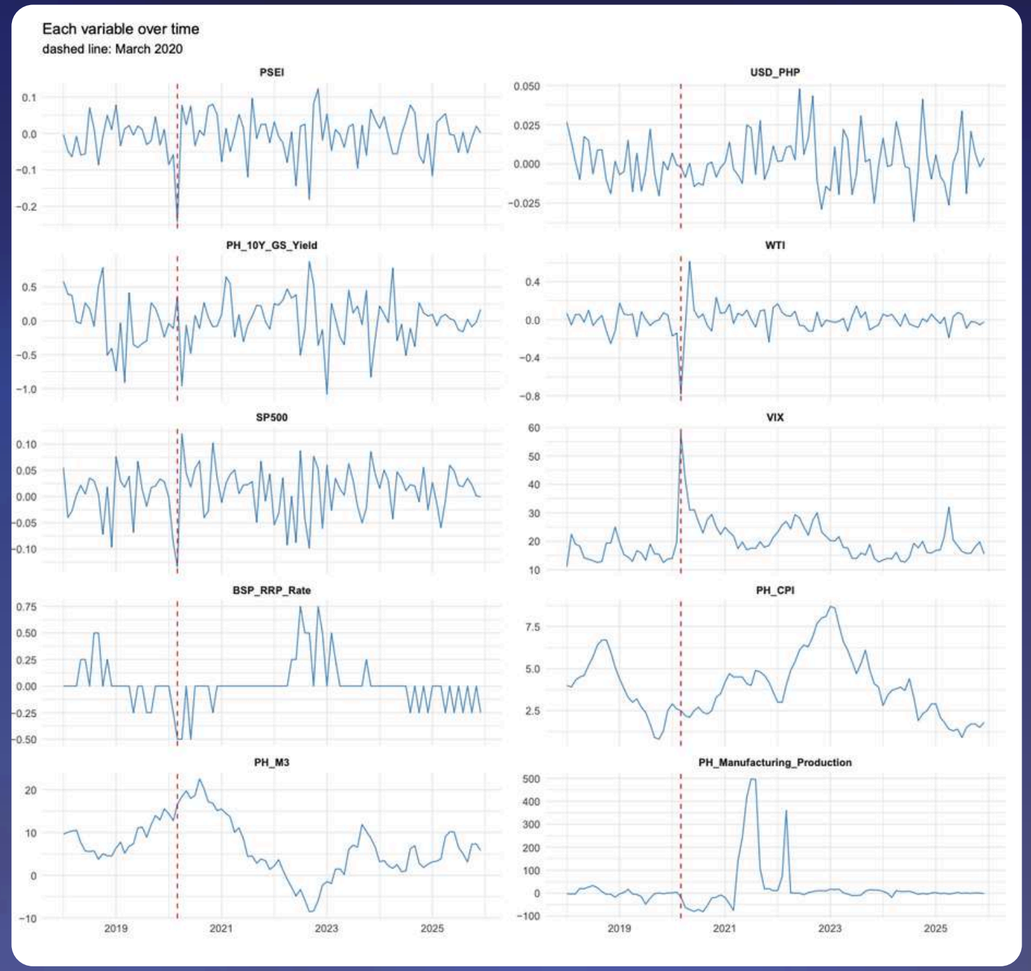
PH_CPI peaked at approximately 8.7 in 2022–2023, followed by monetary tightening.

4. Monetary Policy Adjustments

BSP initially reduced its policy rate during the pandemic before implementing subsequent increases.

5. Other Variable Movements

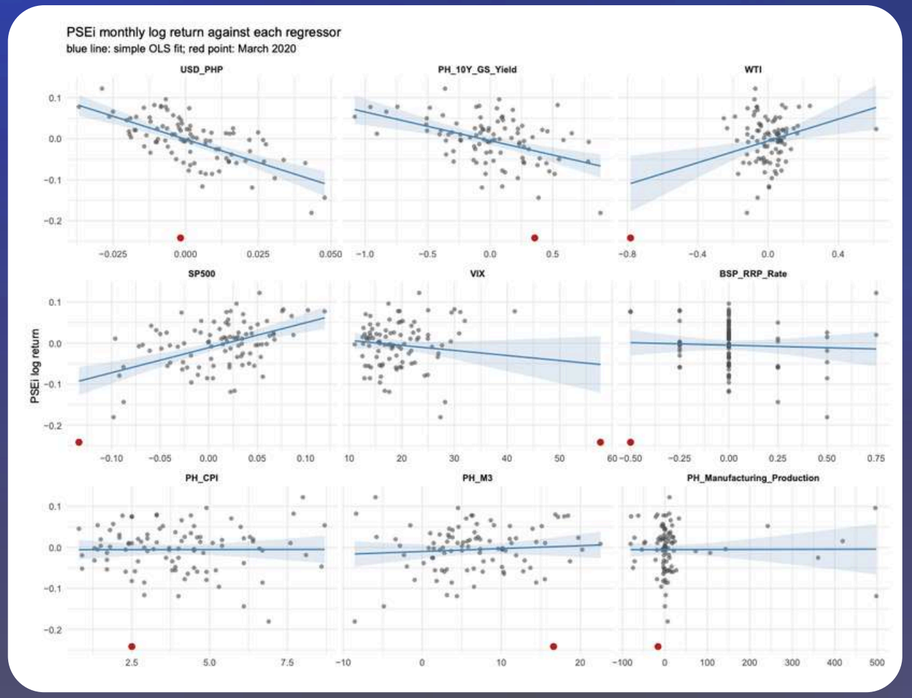
PH_M3, USD_PHP, and PH_10Y_GS_Yield exhibited fluctuations without one clearly identifiable dominant structural break.





EXPLORATORY DATA ANALYSIS

Scatter Plot Analysis



1. Strongest Positive Relationship

SP500 exhibits the clearest positive association with PSEi returns.

2. Negative Relationships

USD_PHP and PH_10Y_GS_Yield show negative associations with PSEi returns.

3. Weak or Uncertain Relationships

WTI and VIX show uncertain associations, while PH_CPI, PH_M3, PH_Manufacturing_Production, and BSP_RRP_Rate exhibit nearly flat fitted lines.

4. Potentially Influential COVID-19 Observation

March 2020 observations fall substantially below fitted lines and may affect estimated relationships.



EXPLORATORY DATA ANALYSIS

Pearson Correlation Analysis

1. **USD_PHP has the strongest correlation with PSEi returns.**
   - *USD_PHP exhibits the strongest negative correlation (r = -0.600).*
   - *SP500 shows a moderate positive correlation (r = 0.487).*
   - *PH_10Y_GS_Yield shows a moderate negative correlation (r = -0.432).*
   - *WTI has a weaker positive correlation (r = 0.302).*

2. **Five explanatory variables have no significant correlation with PSEi returns.**
   - *VIX*
   - *PH_M3*
   - *BSP_RRP_Rate*
   - *PH_CPI*
   - *PH_Manufacturing_Production*

These findings are consistent with the relatively flat fitted regression lines observed in the scatter plots.

3. **Some explanatory variables are correlated with one another.**
   - *BSP_RRP_Rate and PH_CPI have the strongest pairwise correlation (r = 0.658).*
   - *PH_CPI and PH_M3 are negatively correlated (r = -0.559).*
   - *BSP_RRP_Rate and PH_M3 are also negatively correlated (r = -0.511).*
   - *SP500 is correlated with several other variables, including WTI, PH_10Y_GS_Yield, USD_PHP, and VIX.*

4. ***Severe multicollinearity appears unlikely based on pairwise correlations.***
   - *No pairwise correlation exceeds 0.70.*
   - *However, overlapping relationships among explanatory variables may affect individual regression coefficients.*
   - *VIF analysis is still necessary to assess potential multicollinearity.*

1. Significant Correlations with PSEi Returns

Variable	Pearson r
USD_PHP	-0.600
SP500	+0.487
PH_10Y_GS_Yield	-0.432
WTI	+0.302

2. Other Notable Correlations

Variable Pair	Pearson r
BSP_RRP_Rate – PH_CPI	+0.658
PH_CPI – PH_M3	-0.559
BSP_RRP_Rate – PH_M3	-0.511



BASELINE MULTIPLE LINEAR REGRESSION

Initial Multiple Regression

term	estimate	std_error	t_stat	p_value	ci_lower	ci_upper	signif
(Intercept)	0.0065800	0.0192800	0.3412	0.734	-0.0317500	0.0449100	
USD_PHP	-2.0450000	0.3038000	-6.7330	<0.0001	-2.6490000	-1.4410000	***
PH_10Y_GS_Yield	-0.0405200	0.0128100	-3.1650	0.00215	-0.0659800	-0.0150700	***
WTI	0.1297000	0.0364800	3.5550	0.000617	0.0571700	0.2022000	***
SP500	0.1342000	0.1144000	1.1730	0.244	-0.0931600	0.3615000	
VIX	-0.0002430	0.0006980	-0.3482	0.729	-0.0016310	0.0011440	
BSP_RRP_Rate	-0.0120300	0.0271400	-0.4433	0.659	-0.0659800	0.0419200	
PH_CPI	0.0006358	0.0033080	0.1922	0.848	-0.0059400	0.0072120	
PH_M3	-0.0011240	0.0008602	-1.3060	0.195	-0.0028340	0.0005865	
PH_Manufacturing_Production	0.0000330	0.0000446	0.7411	0.461	-0.0000556	0.0001217	

Table 17: Model fit and overall F-test (H0: all slope coefficients = 0)

statistic	value
n	96
r_squared	0.5724
adj_r_squared	0.5277
residual_std_err	0.04102
f_statistic	12.79
f_df_numerator	9
f_df_denominator	86
f_p_value	<0.0001

1. Model Performance

- *The model explains **57.24% of the variation** in PSEi returns (R^2 = 0.5724).*
- *Adjusted R^2 of 0.5277 indicates reasonable explanatory power after accounting for the nine variables.*
- *The model is statistically significant overall (F = 12.79, p < 0.0001).*
- *The nine explanatory variables are jointly significant in explaining PSEi returns.*

2. Statistically Significant Variables

- *Three variables are statistically significant at the 1% level:*
  - ***USD_PHP (β = -2.045, p < 0.0001): P**eso depreciation is associated with lower PSEi returns, holding other variables constant.*
  - ***PH_10Y_GS_Yield (β = -0.0405, p = 0.00215):** Higher government bond yields are associated with lower PSEi returns, potentially reflecting competition between fixed-income and equity investments.*
  - ***WTI (β = 0.1297, p = 0.000617):** Higher oil returns are associated with higher PSEi returns. The result warrants further investigation due to the potential influence of the March 2020 observation.*

3. Statistically Insignificant Variables

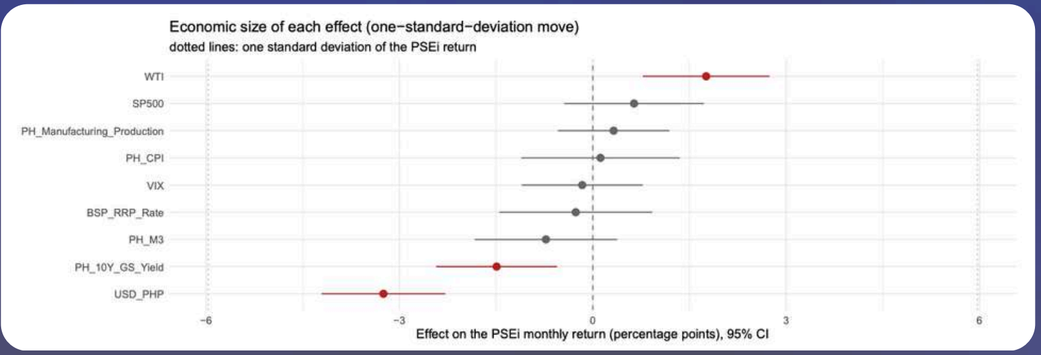
- *Six variables are not statistically significant, even at the 10% level:*
  - *SP500, VIX, BSP_RRP_Rate, PH_CPI, PH_M3, PH_Manufacturing_Production*
    - ***SP500:** Previously showed a significant positive correlation with PSEi returns (r = 0.487), but became insignificant in the MLR model (p = 0.244).*
    - ***PH_Manufacturing_Production**: Remains insignificant (p = 0.461), although its extreme outliers may influence the estimated coefficient.*



BASELINE MULTIPLE LINEAR REGRESSION

Economic Significance

term	sd_of_regressor	slope	effect_pp	ci_low_pp	ci_high_pp	std_beta	p_value	significant
USD_PHP	0.0159	-2.05e+00	-3.250	-4.210	-2.290	-0.5440	<0.0001	yes
PH_10Y_GS_Yield	0.3680	-4.05e-02	-1.490	-2.430	-0.555	-0.2500	0.00215	yes
WTI	0.1360	1.30e-01	1.760	0.775	2.740	0.2950	0.000617	yes
SP500	0.0477	1.34e-01	0.640	-0.444	1.720	0.1070	0.244	no
VIX	6.7600	-2.43e-04	-0.164	-1.100	0.774	-0.0275	0.729	no
BSP_RRP_Rate	0.2200	-1.20e-02	-0.265	-1.450	0.923	-0.0444	0.659	no
PH_CPI	1.8700	6.36e-04	0.119	-1.110	1.350	0.0199	0.848	no
PH_M3	6.4700	-1.12e-03	-0.727	-1.830	0.380	-0.1220	0.195	no
PH_Manufacturing_Production	97.7000	3.31e-05	0.323	-0.543	1.190	0.0541	0.461	no



USD_PHP has the largest estimated economic impact

- *A one-standard-deviation increase is associated with a **3.250-percentage-point decrease** in PSEi returns (β* = -0.5440).*

WTI and PH_10Y_GS_Yield show economically meaningful effects.

- ***WTI:** +1.760 percentage points (β* = 0.2950).*
- ***PH_10Y_GS_Yield: -**1.490 percentage points (β* = -0.2500).*
- ***Both have statistically significant coefficients.***

SP500 and PH_M3 have smaller, statistically insignificant effects.

- ***SP500:** +0.640 percentage points (β* = 0.1070).*
- ***PH_M3:** -0.727 percentage points (β* = -0.1220).*

PH_Manufacturing_Producti on has the smallest estimated effect.

- *A one-standard-deviation increase corresponds to only a **0.323-percentage-point increase** in PSEi returns (β* = 0.0541).*



BASELINE MULTIPLE LINEAR REGRESSION

Joint F-tests Analysis

test_group	H0	q	df_resid	F_ classical	p_ classical	p_hc3	p_newey_ west	decision_ classical
All slopes (overall significance)	b = 0 for all 9 coefficients	9	86	12.790	<0.0001	<0.0001	<0.0001	Reject H0
Global factors: SP500, WTI, VIX	b = 0 for all 3 coefficients	3	86	8.368	<0.0001	0.0901	0.00303	Reject H0
Domestic macro: PH_CPI, PH_M3, PH_ Manufacturing_ Production	b = 0 for all 3 coefficients	3	86	1.070	0.366	0.644	0.355	Fail to reject H0
Monetary and FX: BSP_RRP_Rate, PH_10Y_GS_ Yield, USD_PHP	b = 0 for all 3 coefficients	3	86	21.440	<0.0001	<0.0001	<0.0001	Reject H0
Individually significant three: USD_PHP, PH_ 10Y_GS_Yield, WTI	b = 0 for all 3 coefficients	3	86	21.650	<0.0001	<0.0001	<0.0001	Reject H0

The model is statistically significant.

- All nine explanatory variables are jointly significant ($p < 0.0001$) across classical, HC3, and Newey-West standard errors.

Monetary and foreign exchange factors are strongly significant.

- BSP_RRP_Rate, PH_10Y_GS_Yield, and USD_PHP are jointly significant across all methods ($p < 0.0001$).
- Significance is primarily driven by USD_PHP and PH_10Y_GS_Yield.

Global factors show sensitivity to standard error methods.

- Significant under classical and Newey-West, but only marginally significant under HC3 ($p = 0.0901$).
- WTI appears to be the main contributor to the group's significance.

Domestic macroeconomic factors are not jointly significant.

- PH_CPI, PH_M3, and PH_Manufacturing_Production have p-values above 0.05 across all methods.
- Suggests limited additional explanatory power in the baseline model.

USD_PHP, PH_10Y_GS_Yield, and WTI are jointly significant.

- Remain significant across all three methods ($p < 0.0001$).
- Confirms that their combined relationship with PSEi returns is robust to adjustments for heteroskedasticity and autocorrelation.



MODEL EVALUATION

Candidate Models

model	n	n_regressors	r_squared	adj_r_squared	predicted_r_squared	rmse_in_sample	residual_std_err	f_statistic	f_p_value
M0: intercept only	96	0	0.00000	0.00000	-0.02116	0.05937	0.05968	NA	NA
M1: core 3	96	3	0.54590	0.53110	0.45320	0.04001	0.04087	36.8700	<0.0001
M2: domestic macro	96	4	0.01072	-0.03276	-0.17800	0.05905	0.06065	0.2465	0.911
M3: market-related	96	5	0.55570	0.53100	0.39310	0.03957	0.04087	22.5100	<0.0001
M4: full 9	96	9	0.57240	0.52770	0.33750	0.03882	0.04102	12.7900	<0.0001

Intercept Only (M0)

- *No regressors; baseline model.*
- *$R^2 = 0.0000$.*

Core Regressors (M1)

- *USD/PHP, 10-year yield, WTI.*
- *Highest adjusted $R^2 = 0.5311$.*

Domestic Macro (M2)

- *BSP RRP, CPI, M3, Manufacturing.*
- *Weak fit ($R^2 = 0.0107$).*

Market-related Regressor (M3)

- *Core 3 + S&P 500 and VIX.*
- *$R^2 = 0.5557$.*

Fullset Regressor (M4)

- *All nine regressors.*
- *Highest $R^2 = 0.5724$.*

Overall Model Evaluation

- **Highest R^2:** Full Model (0.5724), indicating the greatest in-sample explanatory power.
- **Highest Adjusted R^2:** Core 3 Model (0.5311), showing the best fit after accounting for complexity.
- **Highest Predicted R^2:** Core 3 Model (0.4532), indicating better leave-one-out performance.
- **Weakest Model:** Domestic Macro 4 (R^2 = 0.0107), demonstrating limited explanatory power.
- **Overall Conclusion:** The Core 3 Model provides the best balance between explanatory power and model complexity, making it the preferred candidate for further evaluation.



MODEL EVALUATION

AIC-BIC

model	k	logLik	AIC	AICc	BIC	delta_AIC	delta_AICc	delta_BIC	weight_AICc	weight_BIC	rank_AIC	rank_BIC
M0: intercept only	2	134.888	-265.776	-265.647	-260.647	69.787	69.249	62.094	0.000	0.000	4	4
M1: core 3	5	172.781	-335.562	-334.896	-322.741	0.000	0.000	0.000	0.771	0.971	1	1
M2: domestic macro	6	135.405	-258.811	-257.867	-243.424	76.752	77.029	79.316	0.000	0.000	5	5
M3: market-related	7	173.827	-333.654	-332.381	-315.703	1.909	2.515	7.037	0.219	0.029	2	2
M4: full 9	11	175.668	-329.335	-326.192	-301.127	6.227	8.704	21.613	0.010	0.000	3	3

criterion	preferred
AIC	M1: core 3
AICc	M1: core 3
BIC	M1: core 3

AIC–BIC Model Evaluation

- **AIC:** Core 3 Model achieved the lowest value (-335.562), indicating the preferred balance between fit and complexity.
- **AICc:** Core 3 Model recorded the lowest value (-334.896), accounting for the limited sample size.
- **BIC:** Core 3 Model achieved the lowest value (-322.741), favoring a simpler specification.
- **Model Support:** Core 3 Model received an AICc weight of 0.771 and BIC weight of 0.971.
- **Overall Conclusion:** All three criteria favor the Core 3 Model (USD/PHP, 10-year bond yield, and WTI), suggesting that additional regressors do not sufficiently improve model fit to justify their complexity.



MODEL EVALUATION

ANOVA F-Test

restricted	full	extra_terms	q	df_resid	RSS_restricted	RSS_full	F	p_value	decision
M0: intercept only	M4: full 9	USD_PHP, PH_10Y_GS_Yield, WTI, SP500, VIX, BSP_RRP_Rate, PH_CPI, PH_M3, PH_Manufacturing_Production	9	86	0.3383	0.1447	12.7900	<0.0001	Reject H0 (keep extra terms)
M1: core 3	M4: full 9	SP500, VIX, BSP_RRP_Rate, PH_CPI, PH_M3, PH_Manufacturing_Production	6	86	0.1536	0.1447	0.8883	0.507	Fail to reject H0 (restriction holds)
M2: domestic macro	M4: full 9	USD_PHP, PH_10Y_GS_Yield, WTI, SP500, VIX	5	86	0.3347	0.1447	22.5900	<0.0001	Reject H0 (keep extra terms)
M3: market-related	M4: full 9	BSP_RRP_Rate, PH_CPI, PH_M3, PH_Manufacturing_Production	4	86	0.1503	0.1447	0.8404	0.503	Fail to reject H0 (restriction holds)
M1: core 3	M3: market-related	SP500, VIX	2	90	0.1536	0.1503	0.9911	0.375	Fail to reject H0 (restriction holds)

ANOVA F-Test Model Evaluation

- **Overall Model Significance:** The full model is statistically significant ($F = 12.79$, $p < 0.0001$).
- **Core 3 vs. Full Model:** Six additional regressors are jointly insignificant ($p = 0.507$).
- **Domestic Macro vs. Full Model:** Adding financial and market-related variables significantly improves model fit ($p < 0.0001$).
- **Market-Related vs. Full Model:** Four additional domestic variables do not significantly improve model fit ($p = 0.503$).
- **Overall Conclusion:** The Core 3 Model captures most of the explanatory power, while additional regressors provide no statistically significant improvement.



MODEL EVALUATION

Wald Test

restricted	full	q	chisq_ classical	p_chisq_ classical	F_ classical	p_F_ classical	F_hc3	p_F_hc3	F_nw	p_F_nw
M0: intercept only	M4: full 9	9	115.100	<0.0001	12.7900	<0.0001	11.0700	<0.0001	22.6400	<0.0001
M1: core 3	M4: full 9	6	5.330	0.502	0.8883	0.507	0.4143	0.868	0.7238	0.632
M2: domestic macro	M4: full 9	5	113.000	<0.0001	22.5900	<0.0001	16.3200	<0.0001	22.1400	<0.0001
M3: market-related	M4: full 9	4	3.362	0.499	0.8404	0.503	0.4203	0.794	0.8371	0.505
M1: core 3	M3: market-related	2	1.982	0.371	0.9911	0.375	0.6844	0.507	0.8627	0.425

Wald Test Model Evaluation

- **Core 3 vs. Full Model (HC3):** Six additional regressors remain jointly insignificant (p = 0.868).
- **Core 3 vs. Full Model (Newey-West):** Confirms the HC3 findings (p = 0.632).
- **Domestic Macro vs. Full Model:** Additional financial and market-related variables significantly improve model fit under both methods (p < 0.0001).
- **Market-Related vs. Full Model:** Four additional domestic variables remain insignificant under HC3 (p = 0.794) and Newey-West (p = 0.505).
- **Overall Conclusion:** Robust Wald tests support the Core 3 Model, confirming that additional regressors do not significantly improve explanatory power.



MODEL EVALUATION

Stepwise Selection

Stepwise Selection Model Evaluation

- **AIC Selection:** Retained USD/PHP, 10-year bond yield, WTI, and M3 money supply growth (Adjusted R^2 = 0.5402).
- **BIC Selection:** Selected only the Core 3 regressors: USD/PHP, 10-year bond yield, and WTI (Adjusted R^2 = 0.5311).
- **Common Variables:** Both methods consistently selected USD/PHP, 10-year bond yield, and WTI.
- **Model Complexity:** AIC favored four regressors, while BIC preferred a simpler three-variable model.
- **Overall Conclusion:** Stepwise selection supports the Core 3 Model, confirming its strong explanatory power with fewer variables.

direction	criterion	selected	n_ regressors	adj_r_ squared	AIC	BIC
both	AIC	USD_PHP, PH_10Y_GS_ Yield, WTI, PH_M3	4	0.5402	-336.5	-321.1
both	BIC	USD_PHP, PH_10Y_GS_ Yield, WTI	3	0.5311	-335.6	-322.7

Step	Df	Deviance	Resid. Df	Resid. Dev	criterion_ value
	NA	NA	86	0.1447	-603.8
- PH_CPI	1	0.0000621	87	0.1447	-605.7
- VIX	1	0.0001688	88	0.1449	-607.6
- BSP_RRP_Rate	1	0.0002419	89	0.1451	-609.5
- PH_Manufacturing_Production	1	0.0010880	90	0.1462	-610.7
- SP500	1	0.0027970	91	0.1490	-610.9

Step	Df	Deviance	Resid. Df	Resid. Dev	criterion_ value
	NA	NA	86	0.1447	-578.1
- PH_CPI	1	0.0000621	87	0.1447	-582.7
- VIX	1	0.0001688	88	0.1449	-587.1
- BSP_RRP_Rate	1	0.0002419	89	0.1451	-591.5
- PH_Manufacturing_Production	1	0.0010880	90	0.1462	-595.4
- SP500	1	0.0027970	91	0.1490	-598.1
- PH_M3	1	0.0046080	92	0.1536	-599.7



REGRESSION DIAGNOSTICS

Heteroskedacity

1. Initial Breusch-Pagan Test

- The test produced a p-value of 0.0254, which is below the 0.05 significance level.
- This means we rejected the null hypothesis of constant error variance, indicating possible heteroskedasticity in the initial model.

2. White Test Results

- The White tests produced p-values of 0.125 and 0.487.
- Both were above 0.05, meaning there was not enough evidence to conclude that heteroskedasticity was present. However, the initial tests gave mixed results.

3. Effect of the March 2020 Dummy

- After adding a dummy variable for the COVID-19 market crash in March 2020, the Breusch-Pagan p-value increased to 0.129.
- This suggests that the unusual market movement during the pandemic contributed to the initial heteroskedasticity issue.

4. Final Model and Adjustment

- The reduced model further increased the Breusch-Pagan p-value to 0.33, meaning we failed to reject the null hypothesis.
- HC3 robust standard errors were also applied to make the statistical results more reliable even when error variance is not constant.

Table 32: Heteroskedasticity tests (H0: homoskedastic errors)

test	statistic	df	p_value	decision
Breusch-Pagan (studentized / Koenker)	18.970	9	0.0254	Reject H0
Breusch-Pagan (non-studentized)	21.670	9	0.01	Reject H0
White-type (fitted, fitted^2)	4.163	2	0.125	Fail to reject H0
White (regressors, squares, cross-products)	53.670	54	0.487	Fail to reject H0

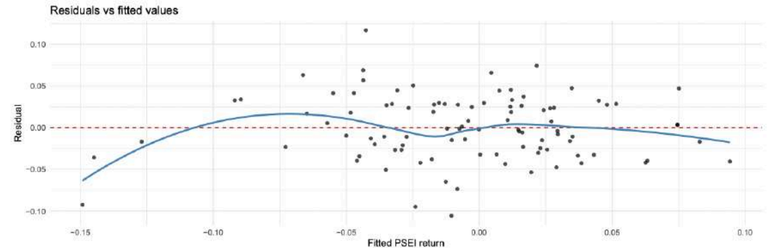


Figure 8: Initial model: residuals against fitted values

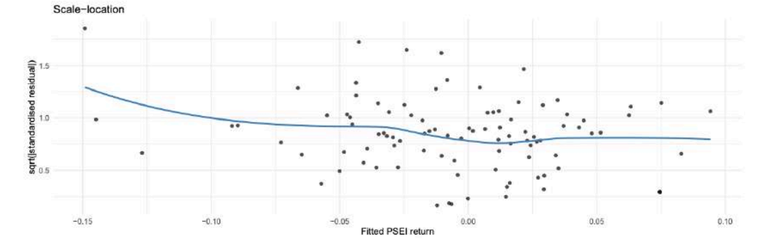


Figure 9: Initial model: scale-location plot, square root of the absolute standardised residuals against fitted values



REGRESSION DIAGNOSTICS

Autocorellation

1. Durbin-Watson Test

- The test produced a statistic of 2.105, which is close to 2, indicating little to no first-order autocorrelation.
- The p-value of 0.896 shows no significant evidence of autocorrelation in either direction.

2. Breusch-Godfrey Test

- Tests at lags 1, 3, 6, and 12 produced p-values of 0.484, 0.794, 0.445, and 0.425, respectively.
- All values are above 0.05, meaning we failed to reject the null hypothesis of no autocorrelation.

3. Ljung-Box Test

- The tests at 6 and 12 lags produced p-values of 0.556 and 0.692.
- This indicates that there is no significant relationship between the residuals across different months.

4. Overall Result

- **The residual plots also showed no clear pattern over time, with all** 24 autocorrelation values within the 95% confidence bounds.
- Therefore, the model does not show evidence of autocorrelation, and no additional correction was needed.

Table 33: Autocorrelation tests (H0: no serial correlation). DW = 2.105; implied lag-1 correlation ≈ -0.052

test	statistic	df	p_value	decision
Durbin-Watson (H1: positive autocorrelation)	2.1050		0.552	Fail to reject H0
Durbin-Watson (H1: two-sided)	2.1050		0.896	Fail to reject H0
Breusch-Godfrey, order 1	0.4906	1	0.484	Fail to reject H0
Breusch-Godfrey, order 3	1.0310	3	0.794	Fail to reject H0
Breusch-Godfrey, order 6	5.8060	6	0.445	Fail to reject H0
Breusch-Godfrey, order 12	12.2600	12	0.425	Fail to reject H0
Ljung-Box, 6 lags (residuals)	4.9070	6	0.556	Fail to reject H0
Ljung-Box, 12 lags (residuals)	9.1320	12	0.692	Fail to reject H0

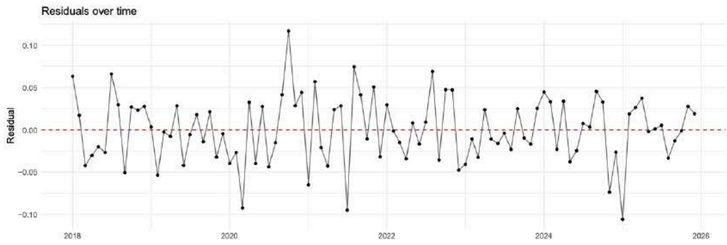
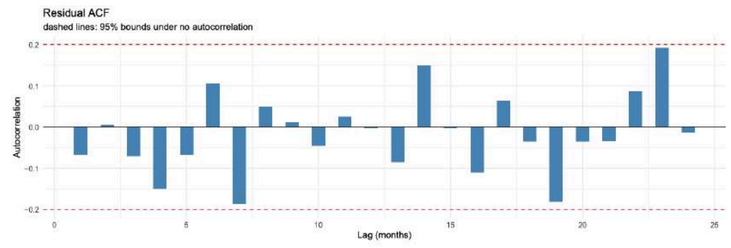


Figure 10: Initial model: residuals in month order





REGRESSION DIAGNOSTICS

Non-Normality

1. Shapiro-Wilk Test
- The test produced a p-value of 0.421, which is above the 0.05 significance level.
- This means we failed to reject the null hypothesis, indicating that the residuals are normally distributed.

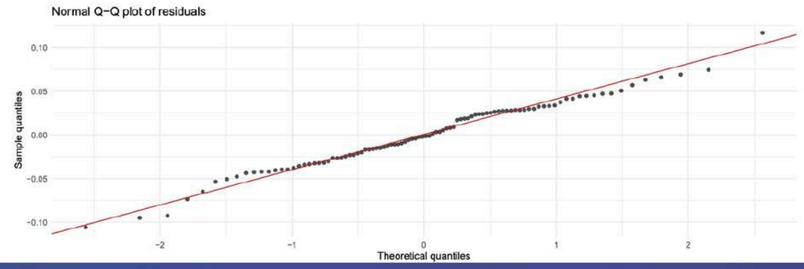
2. Jarque-Bera Test
- The test resulted in a p-value of 0.797, also above 0.05.
- This supports the finding that there is no significant evidence of non-normality in the residuals.

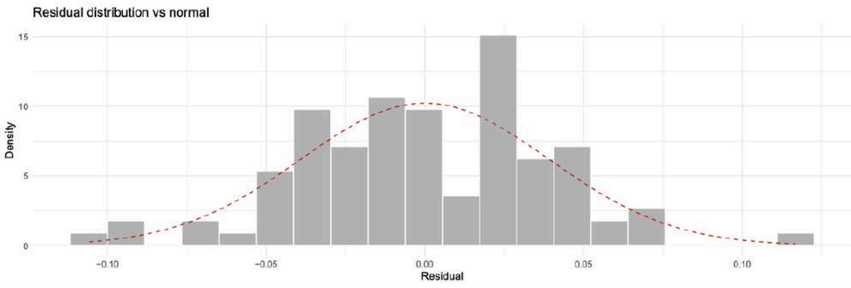
3. Normal Q-Q Plot
- The residuals generally follow the reference line, with only slight deviations at both ends.
- This suggests that the residuals are reasonably close to a normal distribution.

4. Overall Result
- Both statistical tests and the Q-Q plot support the assumption of normality.
- Therefore, no additional correction or adjustment was needed for non-normality.

Table 34: Normality tests on residuals (skewness = -0.091, excess kurtosis = 0.284)

test	statistic	df	p_value	decision
Shapiro-Wilk	0.9863		0.421	Fail to reject H0
Jarque-Bera	0.4545	2	0.797	Fail to reject H0







REGRESSION DIAGNOSTICS

Functional Form (Ramsey Reset) & Multicollinearity

1. Ramsey RESET Test

- The initial model produced a p-value of 0.0201, below the 0.05 significance level.
- This indicates that the model may have a functional form problem or may be missing important terms.

2. Adjustment to the Functional Form

- A dummy variable for March 2020 was added to account for the unusual market crash during the pandemic.
- The p-value increased to 0.573, suggesting that the functional form issue was addressed.

3. Variance Inflation Factor (VIF)

- The average VIF was 1.54, while the highest was 2.16, both below the common threshold of 5.
- This indicates that the independent variables do not have problematic levels of correlation with one another.

4. Overall Result

- The condition number was 2.89, well below the warning level of 30.
- Therefore, the model showed no serious multicollinearity, while the RESET results improved after adding the March 2020 dummy.

Table 35: Ramsey RESET specification tests (H0: correct functional form)

test	statistic	df	p_value	decision
RESET: fitted^2	4.346	1, 85	0.0401	Reject H0
RESET: fitted^2, fitted^3	4.093	2, 84	0.0201	Reject H0
RESET: fitted^2 to fitted^4	5.407	3, 83	0.0019	Reject H0
RESET: squares of regressors	1.289	9, 77	0.257	Fail to reject H0

Table 36: Variance inflation factors (mean VIF = 1.544)

variable	VIF	tolerance	R2_on_others	flag
PH_CPI	2.164	0.462	0.538	low (< 5)
BSP_RRP_Rate	2.014	0.496	0.504	low (< 5)
PH_M3	1.751	0.571	0.429	low (< 5)
SP500	1.680	0.595	0.405	low (< 5)
WTI	1.382	0.723	0.277	low (< 5)
USD_PHP	1.315	0.761	0.239	low (< 5)
VIX	1.258	0.795	0.205	low (< 5)
PH_10Y_GS_Yield	1.256	0.796	0.204	low (< 5)
PH_Manufacturing_Production	1.072	0.933	0.067	low (< 5)

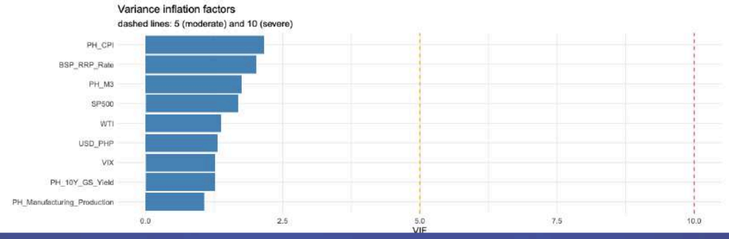


Table 37: Condition number of the standardised regressors

statistic	value
Largest / smallest singular value (condition number)	2.89



REGRESSION DIAGNOSTICS

Influential Observations

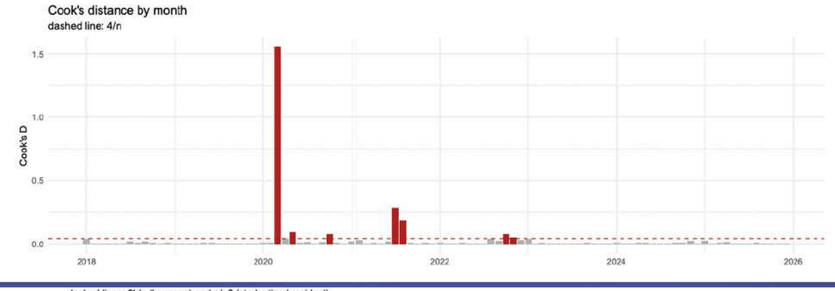
- March 2020 stood out with a Cook's Distance of 1.55, much higher than the other months.
- It also had high leverage (0.57) and a studentized residual of -3.67, making it the most influential observation.
- This was mainly due to the unusual market conditions during COVID-19, when global financial indicators experienced extreme movements.
- Other months, such as July and August 2021, were also flagged but had much smaller influence.

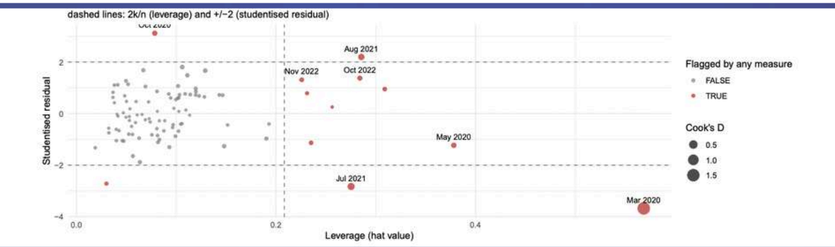
Table 38: Influence measures: cut-offs and number of flagged months

measure	cutoff	n_flagged	max_value
Cook's distance	0.04167	7	1.5530
Leverage	0.20830	10	0.5686
|Studentised residual|	2.00000	5	3.6740
|DFFITS|	0.64550	7	4.2180
Any |DFBETAS|	0.20410	22	2.3260

Table 39: Ten most influential months (by Cook's distance)

year_month	cooks_d	leverage	rstudent	dffits	n_dfbetas_flagged	flag_cook	flag_hat	flag_rstu	flag_dffits	n_flags
2020-03-01	1.5500	0.5690	-3.67	-4.220	7	TRUE	TRUE	TRUE	TRUE	4
2021-07-01	0.2810	0.2750	-2.83	-1.740	2	TRUE	TRUE	TRUE	TRUE	4
2021-08-01	0.1840	0.2860	2.19	1.390	2	TRUE	TRUE	TRUE	TRUE	4
2020-05-01	0.0916	0.3780	-1.23	-0.960	3	TRUE	TRUE	FALSE	TRUE	3
2020-10-01	0.0750	0.0785	3.11	0.909	3	TRUE	FALSE	TRUE	TRUE	3
2022-10-01	0.0741	0.2840	1.37	0.865	7	TRUE	TRUE	FALSE	TRUE	3
2022-11-01	0.0497	0.2260	1.31	0.708	2	TRUE	TRUE	FALSE	TRUE	3
2020-04-01	0.0404	0.3090	0.95	0.635	3	FALSE	TRUE	FALSE	FALSE	1
2018-01-01	0.0403	0.1290	1.67	0.641	4	FALSE	FALSE	FALSE	FALSE	0
2023-01-01	0.0393	0.2350	-1.13	-0.628	3	FALSE	TRUE	FALSE	FALSE	1







REGRESSION DIAGNOSTICS

Parameter Stability

- **OLS-CUSUM Test:** p-value of 0.988, indicating no evidence of unstable regression coefficients.
- **Recursive CUSUM Test:** p-value of 0.206, which also suggests that the model's coefficients remained stable over time.
- **Chow Tests:** All tested break dates had p-values above 0.05, including March 2020 (0.107), December 2021 (0.815), and April 2022 (0.919).
- **Sup-F Test:** The estimated break date was August 2020, but its p-value of 0.306 was not statistically significant.

Overall Result: The tests failed to reject the null hypothesis of parameter stability. This suggests that there is no strong evidence of a structural break in the model during the period studied.

Table 41: Stability tests (H0: stable coefficients)

test	statistic	df	p_value	decision
OLS-CUSUM	0.5089		0.958	Fail to reject H0
Recursive CUSUM	0.7022		0.246	Fail to reject H0
Chow: after Mar 2020 (COVID-19 shock)	1.6550		0.107	Fail to reject H0
Chow: after Dec 2021 (sample midpoint)	0.5490		0.85	Fail to reject H0
Chow: after Apr 2022 (start of BSP tightening)	0.4456		0.919	Fail to reject H0
sup-F (unknown break date)	20.1400		0.306	Fail to reject H0

Table 42: Estimated break (first segment ends in this month)

statistic	value
Most likely break date from the sup-F search	2020-08



term	estimate	se_classical	se_hc3	se_ratio	t_hc3	p_classical	p_hc3	ci95_low	ci95_high	decision_changes
(Intercept)	6.58e-03	1.93e-02	3.28e-02	1.700	0.200	0.734	0.842	-0.058700	0.071900	FALSE
USD_PHP	-2.05e+00	3.04e-01	2.95e-01	0.971	-6.940	<0.0001	<0.0001	-2.630000	-1.460000	FALSE
PH_10Y_GS_Yield	-4.05e-02	1.28e-02	1.39e-02	1.090	-2.920	0.00215	0.00453	-0.068200	-0.012900	FALSE
WTI	1.30e-01	3.65e-02	8.86e-02	2.430	1.460	0.000617	0.147	-0.046400	0.306000	TRUE
SP500	1.34e-01	1.14e-01	1.20e-01	1.050	1.120	0.244	0.267	-0.104000	0.373000	FALSE
VIX	-2.43e-04	6.98e-04	1.47e-03	2.100	-0.166	0.729	0.869	-0.003160	0.002670	FALSE
BSP_RRP_Rate	-1.20e-02	2.71e-02	3.12e-02	1.150	-0.385	0.659	0.701	-0.074100	0.050000	FALSE
PH_CPI	6.36e-04	3.31e-03	3.22e-03	0.973	0.197	0.848	0.844	-0.005770	0.007040	FALSE
PH_M3	-1.12e-03	8.60e-04	1.06e-03	1.230	-1.060	0.195	0.292	-0.003230	0.000984	FALSE
PH_Manufacturing_Production	3.31e-05	4.46e-05	9.34e-05	2.090	0.354	0.461	0.724	-0.000153	0.000219	FALSE

term	SE ratio HC3 / classical	SE ratio NW (auto) / classical	sig_classical	sig_hc3	sig_nw_auto	sig_nw_lag6
BSP_RRP_Rate	1.150	0.756	FALSE	FALSE	FALSE	FALSE
PH_10Y_GS_Yield	1.085	0.785	TRUE	TRUE	TRUE	TRUE
PH_CPI	0.973	0.783	FALSE	FALSE	FALSE	FALSE
PH_M3	1.233	1.042	FALSE	FALSE	FALSE	FALSE
PH_Manufacturing_Production	2.094	0.737	FALSE	FALSE	FALSE	FALSE
SP500	1.049	0.983	FALSE	FALSE	FALSE	FALSE
USD_PHP	0.971	0.780	TRUE	TRUE	TRUE	TRUE
VIX	2.101	0.940	FALSE	FALSE	FALSE	FALSE
WTI	2.428	1.016	TRUE	FALSE	TRUE	TRUE

MODEL IMPROVEMENT

Robust Standard Errors (HC3 Estimator)

Addressing Heteroskedasticity

- Initial OLS model exhibited heteroskedasticity (Breusch-Pagan p = 0.025).
- HC3 robust standard errors were applied to improve the reliability of statistical inference.
- Regression coefficients and R^2 remain unchanged; only standard errors, test statistics, and confidence intervals are adjusted.

Statistically Significant Variables

- **USD/PHP (HC3 p < 0.0001):** Remains highly significant, indicating a robust negative association between peso depreciation and PSEi returns.
- **PH 10-Year Government Bond Yield (HC3 p = 0.00453):** Remains significant, supporting the negative association between rising bond yields and PSEi returns.
- **WTI Crude Oil (p = 0.000617 → 0.147):** Becomes statistically insignificant after HC3 correction, suggesting its initial significance was sensitive to the standard-error estimator.

Model Implications

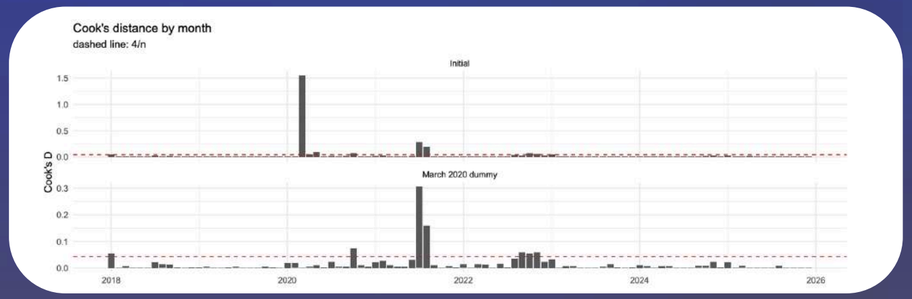
- **Overall Model:** Remains jointly significant using HC3 robust inference (F = 11.1).
- **Reliable Indicators:** USD/PHP and 10-year bond yields maintain statistical significance under the correction.
- **Key Limitation:** HC3 corrects statistical inference for heteroskedasticity but does not eliminate outlier influence or improve the model's fit.

Key Financial Insight

Exchange rates and bond yields remain statistically significant, while WTI's relationship with PSEi returns becomes less reliable after HC3 correction.



term	est_initial	p_initial	est_march	p_march	pct_change	sig_initial	sig_march
(Intercept)	6.58e-03	0.842	-2.07e-02	0.272	-414.0	FALSE	FALSE
USD_PHP	-2.05e+00	<0.0001	-1.98e+00	<0.0001	3.4	TRUE	TRUE
PH_10Y_GS_Yield	-4.05e-02	0.00453	-3.46e-02	0.00513	14.6	TRUE	TRUE
WTI	1.30e-01	0.147	5.04e-02	0.171	-61.1	FALSE	FALSE
SP500	1.34e-01	0.267	1.48e-01	0.193	10.1	FALSE	FALSE
VIX	-2.43e-04	0.869	1.04e-03	0.161	530.0	FALSE	FALSE
BSP_RRP_Rate	-1.20e-02	0.701	-2.32e-02	0.379	-93.0	FALSE	FALSE
PH_CPI	6.36e-04	0.844	9.62e-04	0.751	51.3	FALSE	FALSE
PH_M3	-1.12e-03	0.292	-7.42e-04	0.448	33.9	FALSE	FALSE
PH_Manufacturing_Production	3.31e-05	0.724	3.84e-05	0.679	16.2	FALSE	FALSE



MODEL IMPROVEMENT

March 2020 Dummy Solution

Addressing Influential Observations

- March 2020 COVID-19 crash was highly influential (Cook's D = 1.55).
- A dummy variable was introduced to account for the extreme market shock without removing the observation.
- This reduces the distortion of regression estimates caused by March 2020.

Model Performance Improvement

- **March 2020 Dummy (p = 0.0004):** Statistically significant, confirming the exceptional impact of the COVID-19 market shock.
- **Adjusted R^2 (0.5277 → 0.588):** Improved explanatory power, indicating that the model explains more variation in PSEi returns.
- **Maximum Cook's D (1.55 → 0.31):** Reduced the influence of extreme observations on regression estimates.
- **RESET p-value (0.020 → 0.573):** No longer indicates statistically significant evidence of model misspecification at the 5% significance level.

Model Implications

- **Improved Reliability:** Accounting for the COVID-19 shock strengthens model estimates.
- **WTI Coefficient:** Decreases by 61%, indicating sensitivity to March 2020.
- **Key Limitation:** Other regression assumption violations may remain.

Key Financial Insight

Accounting for extraordinary market shocks improves the reliability of macro-financial relationships used in investment decisions.



term	OLS	p_OLS_hc3	Huber	p_Huber	MM	p_MM
(Intercept)	6.58e-03	0.842	2.38e-03	0.895	-0.015500	0.398
USD_PHP	-2.05e+00	<0.0001	-2.00e+00	<0.0001	-1.940000	<0.0001
PH_10Y_GS_Yield	-4.05e-02	0.00453	-3.87e-02	0.00164	-0.035800	0.00399
WTI	1.30e-01	0.147	1.04e-01	0.00279	0.054000	0.121
SP500	1.34e-01	0.267	1.59e-01	0.138	0.155000	0.154
VIX	-2.43e-04	0.869	1.63e-04	0.802	0.001020	0.127
BSP_RRP_Rate	-1.20e-02	0.701	-2.03e-02	0.422	-0.025100	0.332
PH_CPI	6.36e-04	0.844	9.57e-05	0.975	0.000150	0.962
PH_M3	-1.12e-03	0.292	-1.33e-03	0.101	-0.000968	0.237
PH_Manufacturing_Production	3.31e-05	0.724	6.01e-05	0.15	0.000080	0.0612

year_month	residual_ols	weight_huber	weight_mm
2020-03-01	-0.0925	0.406	0.000
2021-07-01	-0.0949	0.475	0.288
2025-01-01	-0.1060	0.485	0.408
2020-10-01	0.1170	0.464	0.518
2024-11-01	-0.0736	0.680	0.672
2018-01-01	0.0631	0.797	0.702
2021-01-01	-0.0649	0.825	0.718
2018-07-01	0.0659	0.792	0.734
2022-08-01	0.0690	0.762	0.747
2021-02-01	0.0570	0.862	0.782

MODEL IMPROVEMENT

Robust Regression

Addressing Influential Observations

- Extreme market movements may disproportionately affect OLS regression estimates.
- Huber and MM robust regression reduce the influence of outliers without manually excluding observations.
- March 2020 receives a weight of 0.406 under Huber and 0.000 under MM.

Model Performance Improvement

- **Residual Scale:** Decreases from 0.0410 (OLS) to 0.0386 (Huber) and 0.0374 (MM), indicating reduced outlier sensitivity.
- **USD/PHP:** Remains highly significant ($p < 0.0001$), with a consistent negative association.
- **10-Year Bond Yield:** Remains significant under Huber ($p = 0.00164$) and MM ($p = 0.00399$), supporting its negative relationship with PSEi returns.
- **WTI:** Significant under Huber ($p = 0.00279$) but not MM ($p = 0.121$), indicating outlier sensitivity.

Model Implications

- **Improved Robustness:** Reducing outlier influence produces less outlier-sensitive estimates.
- **Consistent Indicators:** Exchange rates and bond yields maintain stable negative relationships with PSEi returns.
- **Key Limitation:** WTI's statistical significance depends on how strongly extreme observations are downweighted.

Key Financial Insight

Exchange rates and bond yields remain consistent indicators of PSEi returns, while oil price relationships are sensitive to extreme market conditions.



term	estimate	se_ classical	se_hc3	t_hc3	p_ classical	p_hc3	ci95_low	ci95_high
(Intercept)	0.000429	0.00396	0.00400	0.107	0.914	0.915	-0.00752	0.00837
USD_PHP	-2.040000	0.25800	0.24200	-8.440	<0.0001	<0.0001	-2.52000	-1.56000
PH_10Y_GS_Yield	-0.040800	0.01120	0.00965	-4.230	0.000458	<0.0001	-0.06000	-0.02170
WTI	0.066700	0.03620	0.03430	1.940	0.0688	0.055	-0.00147	0.13500
MARCH_2020	-0.179000	0.04830	NA	NA	0.000364	NA	NA	NA

Table 54: Can the six regressors be dropped?

test	F	p_value
Nested F-test: drop the six insignificant regressors (H0: all zero)	0.9827	0.442

MODEL IMPROVEMENT

Reduced Model

Addressing Model Complexity

- The initial OLS model included nine explanatory variables, several of which were statistically insignificant.
- A reduced model was developed to eliminate unnecessary predictors and improve model efficiency.
- The final model retained USD/PHP, 10-year government bond yield, WTI, and the March 2020 dummy variable.

Model Performance Improvement

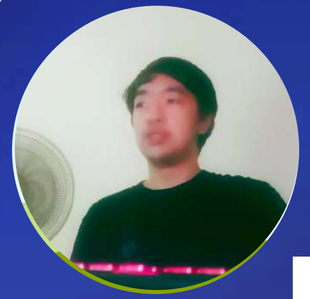
- **Adjusted R^2 (0.5277 → 0.5881):** Improved explanatory power despite using fewer variables.
- **Cross-Validated RMSE (0.04936 → 0.04280):** Lower prediction error, indicating better validation performance.
- **Breusch-Pagan (p = 0.331):** No significant evidence of heteroskedasticity.
- **RESET (p = 0.603):** No significant evidence of functional-form misspecification.
- **Maximum Cook's D (1.55 → 0.059):** Substantially reduced influence of extreme observations.

Model Implications

- **Improved Efficiency:** Achieves stronger model performance with fewer explanatory variables.
- **Better Diagnostics:** Addresses major weaknesses identified in the initial OLS model.
- **Key Limitation:** Results remain based on contemporaneous associations rather than forecasts.

Key Financial Insight

The reduced model provides a simpler and more reliable framework for monitoring PSEi returns through exchange rates, bond yields, and oil prices while accounting for the COVID-19 shock.



LOGISTIC REGRESSION

Multiple Linear Regression (MLR) explains the magnitude of monthly PSEi returns, but it does not directly determine the probability of experiencing a positive or negative market outcome. **Logistic regression addresses this limitation by estimating the likelihood of specific market events.**

The method is particularly relevant because **PSEi returns are left-skewed,** indicating the presence of extreme negative returns that deserve separate examination. Unlike MLR, logistic regression is **designed for binary outcomes and produces probabilities ranging from 0 to 1,** making it suitable for assessing the likelihood of particular market conditions.

The study examines two outcomes through separate models:

MODEL A

Model A estimates the probability of an up month, defined as a month with positive PSEi returns.

MODEL B

Model B estimates the probability of a crash month, defined as a monthly return below -5%.

Firth penalization is applied to Model B to address potential estimation bias caused by the limited number of crash observations. Since only 21 out of the 96 months qualified as crash months, this approach helps improve the reliability of the model's estimates.



LOGISTIC REGRESSION

MODEL A: UP-MONTH LOGIT

- Examines the likelihood of the PSEi recording a positive monthly return.
- 48 out of 96 months (50%) were classified as up months.
- USD/PHP and the Philippine 10-year government bond yield were statistically significant.
- A weaker peso and higher bond yields reduce the odds of an up month.

MODEL B: CRASH-MONTH LOGIT

- Examines the likelihood of experiencing a significant monthly market decline.
- 21 out of 96 months (21.9%) were classified as crash months.
- Uses Firth penalized likelihood to address limited crash observations.
- USD/PHP, 10-year government bond yield, and WTI were statistically significant.
- Peso depreciation and higher yields increase crash odds, while higher oil prices reduce them in the estimated model.



CROSS-VALIDATION AND OUT-OF-TIME (OOT) CHECK

It determines whether the logistic models maintain their performance when tested on observations not used during estimation.

- **Cross-validation:** Repeated stratified 10-fold cross-validation with 30 repetitions.
- **Out-of-time testing:** Models estimated using 2018–2022 data (60 months) and tested on 2023–2025 data (36 months).

- Cross-validated AUC ranged from 0.797 to 0.865.
- OOT AUC ranged from 0.827 to 0.947.
- All models performed better than their respective majority-class benchmarks in OOT accuracy.

Important limitation: *Model B's OOT results are indicative only because the test period contains just nine crash months*





LOGISTIC REGRESSION

Key indicator	Model A: Up Month	Model B: Crash Month
Observations / events	96 / 48	96 / 21
McFadden R^2	0.33	0.42
In-sample AUC	0.87	0.91
Significant variables	USD/PHP, 10Y yield	USD/PHP, 10Y yield, WTI

Key odds ratio findings:

- **USD/PHP (+1%):**
  - **Model A:** Up-month odds decrease by approximately 61%.
  - **Model B:** Crash-month odds increase by approximately 165%.
- **10Y yield (+10 bps):**
  - **Model A:** Up-month odds decrease by approximately 22%.
  - **Model B:** Crash-month odds increase by approximately 22%.
- **WTI (+5%):**
  - **Model A:** No statistically significant effect.
  - **Model B:** Crash-month odds decrease by approximately 25%.





LOGISTIC REGRESSION

Key Insights

- **Peso depreciation is the strongest indicator:** A weaker peso is consistently associated with lower odds of positive PSEi returns and higher odds of crashes.
- **Bond yields negatively affect equity performance: H**igher government bond yields are associated with reduced up-month odds and increased crash-month odds.
- **Oil prices matter primarily during extreme market declines**: However, the relationship is heavily influenced by March 2020 and is not robust when that month is excluded.
- **Domestic macroeconomic variables have limited explanatory contribution:** CPI, M3, manufacturing production, and BSP policy-rate changes were not statistically significant.
- **Logistic regression complements MLR:** It adds information about the probability of positive and crash months rather than merely explaining the magnitude of returns.
- **The models are explanatory, not predictive:** Lagged regressors lose almost all explanatory power, indicating that the relationships primarily reflect same-month movements.



Financial Interpretation





1. Exchange Rate is a Key Factor

- Peso depreciation is associated with lower PSEi returns and a higher likelihood of market crashes.

2. Government Bond Yields Affect Stock Performance

- Higher bond yields may encourage investors to choose bonds over stocks, putting pressure on equity prices.

3. Other Variables Show Weaker Results

- WTI oil prices were not consistently significant, while CPI, M3, manufacturing growth, and BSP policy rates showed no significant independent relationship.

4. Investment Recommendation

- Investors should monitor exchange rates and bond yields, maintain a diversified portfolio, and consider company fundamentals before making investment decisions.



RECOMMENDATIONS

MONITOR USD/PHP EXCHANGE RATE

- **Strongest Negative Signal:** Peso depreciation correlates directly with PSEi declines and elevated market crash probability.
- **Risk Assessment:** Review exposure to foreign-currency liabilities, imported input costs, and foreign capital dependence.

TRACK 10-YEAR GOVERNMENT YIELDS

- **Asset Reallocation:** Rising yields increase fixed-income appeal relative to equities, driving capital rotation.
- **Corporate Debt:** Higher yields elevate borrowing costs—scrutinize heavily leveraged firms reliant on external financing.

DUAL SIGNAL PORTFOLIO DEFENSE

- **Compound Risk:** Simultaneous peso depreciation and rising bond yields signal severe unfavorable market conditions.
- **Defensive Action:** Reallocate into fixed-income; prioritize resilient equity balance sheets with stable cash flows.

AVOID CRUDE OIL PREDICTOR BIAS

- **Inconsistent Metric:** Relationship with PSEi broke down during the March 2020 market crash event.
- **Guidance:** Do not treat WTI crude oil price movements as a primary or reliable predictor of market returns.



THANK YOU

Presented by :
MACAYAN, Adrian Denzel
MIRANDA, Lara Stephanie
OLACO, Tiarra Emilia
ROA, Marco Antonio

CASE STUDY PRESENTATION 1
Section: **C02**